CITY UNIVERSITY OF HONG KONG

Department of Electrical Engineering Progress Report 2026/27 Semester A

PART A

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Programme Major	BEng — Computer and Data Engineering
Course	EE4080
Academic Year / Term	2026/27 A
Project Code and Title	KHC-04 Re-evaluating Multimodal LLMs for Influenza Forecasting
Name of Supervisor	KHC - Prof CHAN, Katie
Name of Assessor	YNS - Prof SUN, Yanni
Remark	Software
Number of meetings with supervisor / assessor	_____ / _____
Average weekly project hours	_____ hours/week

PART B

1. Project objectives

Problem statement. An accurate national influenza curve can still have timing or peak-height errors. I am independently reproducing MIFlu and evaluating how model choice changes when peak errors and parameter count accompany average error. The project evaluates accuracy, peak skill, parameter efficiency and statistical evidence, with explicitly different model subsets.

Objectives. (1) Reproduce and explain Table V. (2) Compare 28 retained implementations under common windows. (3) assess peak timing, magnitude and parameter trade-offs. (4) Test the specific frozen-encoder transfer setup. (5) Validate findings across origins and regions before claiming deployment value.

2. Original project planning chart

Table 1. Original planning periods and current evidence

Period	Planned work	Snapshot status
Sep 2026	Read MIFlu and establish data contract	Local data equality checked
Oct 2026	Reproduce Table V	CSV and oracle arithmetic checked
Nov 2026	Benchmark and ablation; progress presentation	28-arm CSV complete; ablation reported 60/60
Nov–Dec 2026	Regional and rolling-origin validation	Not completed in this snapshot
Jan–Apr 2027	Manuscript strengthening and submission decision	Planned, conditional on validation

This is a reconstruction from the supplied plan, annotated at 3 November 2026. No contemporaneously signed September chart was available.

3. Background study

Patching and linear baselines are established [3–5]. Time-LLM, CALF and TimeCMA use different alignment/adaptation designs [6–8]. Chronos, TimesFM, Moirai and Lag-Llama pretrain on numerical series and were not evaluated here [9–12]. Earlier LLM-removal work already questions the benefit of

language pretraining [13], while a 2026 preprint reports conditional benefits under cross-dataset shifts [14]. The contribution here is the specific influenza protocol audit and multi-metric comparison.

Global and societal context. Peak forecasts can inform preparedness and capacity planning, but retrospective national accuracy does not measure clinical benefit. Transparent evaluation can reduce unjustified spending on complex models and expose failure modes. No real-world resource-saving or patient-outcome effect was measured [2,18].

4. Methodology

Table 2. Separate reproduction and benchmark protocols

Item	Reproduction	Benchmark
Seeds	2021–2030 (10)	42/43/44 (3)
Models	MIFlu	13 conventional + 4 LLM-based + 11 frozen encoders
Selection	Validation checkpoint	Validation checkpoint
Scores	7-variable z-space MSE / MAE	7-variable MSE / MAE; ILITOTAL PTE / PIE
Windows	160 / 144 / 144 / 128	160 / 144 / 144 / 128

Three seeds are a compute-budget choice, not a universal venue rule. Reproduction and benchmark estimates use different seed sets. Do not substitute one for the other. Exact upstream protocol defects remain to be checked against a pinned author revision.

We use a chronological split and train-only scaling. Validation selects checkpoints in this implementation. The handoff attributes test-set checkpoint selection and a one-variable loader to upstream code, but a pinned author checkout is absent and the public endpoint could not be opened. Those upstream allegations remain unverified. The notes also show learning-rate differences between code defaults and paper settings, so the comparison must not be described as differing in only two ways.

Peak timing error (PTE) is the absolute difference between predicted and observed peak weeks. Peak intensity error (PIE) is the absolute peak-height error divided by the observed peak, expressed as a percentage. Kramer–Shaman [2] verifies the separate one-week and 25% tolerances. We apply both to mean errors in each (model, L, step) cell. This joint screen and its peak detector are project choices, not a clinical acceptance standard. A mean-error pass is not an individual-season success probability.

Design impact. All retained arms use the same cropped evaluation windows. Scaling is fitted on training data. The 7-variable average is dimensionless, while peak errors use ILITOTAL in patient counts. This allows transparent comparisons without pretending the joint screen certifies safe public-health use. Three benchmark seeds reflect compute limits; additional independent seasons are needed for uncertainty.

6. Progress

Progress against plan. The supplied exports contain the reproduction table and 28 × 4 × 3 benchmark cell means. The original suite is supplied under tests_archive/, but cannot be collected cleanly here. My audit now recomputes the CSV aggregates and has seven new CPU tests. The returned queue reports 60/60 ablation cells DONE; per-variant aggregation remains pending. Regional and rolling-origin evaluation are incomplete.

Execution update (5 November 2026 00:15 HKT). Core artefacts: 160/160 complete. The national 1120 matrix is in progress: packaged ledger 117 completed, later owner report 141. Supplied CPU no-text record: 3 passed, 0 skipped. Ablation queue: 60/60 DONE, aggregate trend pending. These execution records do not replace the historical results below.

A common ten-seed result is not yet available. MIFlu stored metrics use cropped windows, while other supplied models use all valid windows; the complete seven-channel MIFlu prediction tensors were not attached. Scores must be harmonised before a common ranking. The submitted protocol still has frozen:false and no lock timestamp, so these comparisons remain retrospective/exploratory. The two data profiles have different byte hashes; the supplied aligned seven-channel comparison reports maximum numeric difference 0.0, which is not byte identity or independent verification of upstream provenance.

At the 00:15 handoff the owner reports 13 empty decision_tree partial directories and two pending gpu_a100 lanes. Table VI (19 TODO) and walk-forward (170 TODO) have not started in the v2 successor chain, although earlier legacy DONE rows exist. One earlier summary lists 9 regional TODOs; execution-end status must reconcile the dates and skip counts. Storage is owner-reported at approximately 306/300 GB soft, with ten-day grace and 1024 GB hard; the source quota log uses KB units. These are dated reports, not current live readings.

Table 3. MIFlu reproduction compared with the paper

Horizon	Our MSE ± SD	Paper MSE	Δ MSE	Our MAE	Paper MAE	Δ MAE
L=24	1.3045 ± 0.0883	1.5420	-15.4%	0.7482	0.7260	+3.1%
L=36	1.3561 ± 0.1603	1.4220	-4.6%	0.7712	0.7790	-1.0%
L=48	1.6543 ± 0.1064	1.4140	+17.0%	0.8381	0.7570	+10.7%
L=60	1.6829 ± 0.2109	1.3640	+23.4%	0.8596	0.7190	+19.6%
4-horizon mean	1.4995	1.4355	+4.5%	0.8043	0.7452	+7.9%

Source: ESSAY_VS_REPRODUCTION_TABLE_V.csv, lines 2–6. SD is recomputed from the code-transcribed ten-seed oracle; its raw run logs are not included. Positive differences mean higher error.

The reproduction CSV reports mean MSE 1.4995 versus 1.4355, a 4.5% increase. MSE is lower at L24 and L36 and higher at L48 and L60. Mean MAE is 0.8043 versus 0.7452 (+7.9%), but its signs are one lower and three higher; MAE does not share the MSE 2–2 crossover. Protocol and missing upstream provenance limit claims of exact replication.

Reproduction conclusion. The observed mean-MSE gap is +4.5%. The remaining protocol and provenance differences prevent a claim of exact replication or superiority.

Table 4. Accuracy of all retained benchmark arms

#	Model	L24	L36	L48	L60	Mean MSE	Mean MAE
1	MIFlu	1.3323	1.2632	1.7774	1.8567	1.5574	0.8199
2	PatchTST	1.4402	1.6154	1.9288	1.6640	1.6621	0.8786
3	NLinear	1.7971	1.7950	1.9574	1.9368	1.8716	0.9183
4	DNN	1.8920	1.7684	2.2603	2.6451	2.1415	1.0057
5	TimeCMA	2.3281	2.5794	2.7104	2.8146	2.6081	1.0676
6	Gradient Boosting	3.5658	3.6867	3.6533	3.7521	3.6645	1.2521
7	KNN	3.7231	3.7623	3.7740	3.8820	3.7854	1.3293
8	Random Forest	3.5486	3.8251	4.0159	4.1710	3.8901	1.2779
9	iTransformer	3.7656	4.0596	4.0438	4.1088	3.9945	1.3476
10	Decision Tree	4.5340	4.1380	4.2186	4.3884	4.3198	1.3798
11	Hunyuan-1.8B	4.5629	4.5877	4.7111	5.0879	4.7374	1.4755
12	Hunyuan-7B	4.3438	4.6624	4.9201	5.1120	4.7596	1.4685
13	DeepSeek-R1-Qwen-1.5B	4.6012	4.6242	4.9245	4.9963	4.7866	1.5002
14	DeepSeek-R1-Llama-8B	4.4337	4.7715	4.8929	5.2019	4.8250	1.4786
15	GRU	4.2895	5.3099	5.1751	4.7736	4.8870	1.5174
16	DeepSeek-R1-32B	4.4401	4.8565	5.0136	5.4386	4.9372	1.4935
17	LSTM	4.4965	5.1359	5.2088	5.1939	5.0088	1.5303
18	DeepSeek-R1-Qwen-14B	4.4976	4.9164	5.1687	5.4838	5.0166	1.5019
19	DeepSeek-R1-Qwen-7B	4.5551	4.9991	5.2417	5.3272	5.0308	1.5133
20	RNN	4.4222	5.2893	5.3424	5.2157	5.0674	1.5634
21	ContextPilot-8B	4.6094	5.1318	5.2895	5.3065	5.0843	1.5255
22	Hunyuan-4B	4.7383	5.0535	5.2876	5.4474	5.1317	1.5450
23	ContextPilot-14B	4.7674	5.1566	5.2168	5.6388	5.1949	1.5407
24	Qwen2.5-1.5B	4.7840	5.0587	5.4546	5.5224	5.2049	1.5725
25	Qwen-Time	4.7190	5.1325	5.6894	5.4818	5.2557	1.5794
26	SVR	5.2248	5.3581	5.3781	5.3940	5.3388	1.4986
27	CALF	5.3624	5.5113	5.6442	5.7962	5.5785	1.6190
28	CNN	5.5021	5.3098	5.2801	6.3698	5.6155	1.6720

Source: BENCH_MSE_BY_MODEL_L.csv, keyed by model and L. Each row averages three seeds on 160/144/144/128 windows. DLinear is excluded by project choice; that exclusion needs a documented sensitivity analysis. Names denote project implementations, whose equivalence to published methods is not established by this ranking.

F0 conclusion. MIFlu has the lowest observed MSE (1.5574), but MSE alone does not establish public-health utility. The top-three spread is 20.2% relative to MIFlu, not less than 20%.

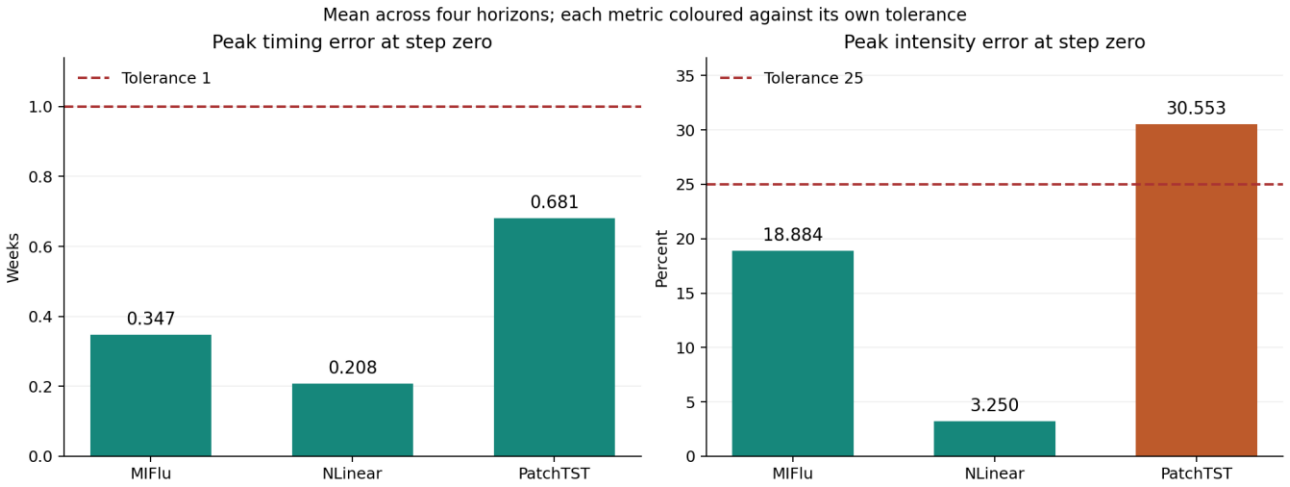
Table 5. Joint peak screen for each horizon and lead

Model	L	s0	s1	s2	s3
NLinear	24	PASS 0.111 / 3.949	fail 1.333 / 8.411	fail 2.333 / 11.943	fail 2.667 / 18.739
NLinear	36	PASS 0.333 / 4.181	fail 1.333 / 7.222	fail 2.111 / 15.313	fail 2.667 / 18.145
NLinear	48	PASS 0.222 / 2.419	fail 1.222 / 4.727	fail 2.222 / 9.979	fail 2.778 / 16.655
NLinear	60	PASS 0.167 / 2.450	fail 1.444 / 4.542	fail 2.556 / 9.952	fail 3.333 / 13.000
MIFlu	24	PASS 0.000 / 18.307	fail 0.889 / 26.686	fail 1.222 / 30.472	fail 1.333 / 31.638
MIFlu	36	PASS 0.000 / 18.695	fail 0.778 / 26.333	fail 1.667 / 31.514	fail 1.333 / 32.427
MIFlu	48	fail 1.222 / 18.004	PASS 0.667 / 24.795	fail 1.222 / 30.133	fail 1.333 / 33.236
MIFlu	60	PASS 0.167 / 20.530	fail 1.333 / 25.799	fail 2.222 / 29.790	fail 2.667 / 31.177
PatchTST	24	fail 0.667 / 30.580	fail 1.000 / 34.044	fail 1.222 / 33.572	fail 1.222 / 30.722
PatchTST	36	fail 1.000 / 26.235	fail 1.667 / 32.247	fail 1.333 / 31.245	fail 1.333 / 33.854
PatchTST	48	fail 0.556 / 30.360	fail 1.333 / 30.878	fail 0.889 / 35.158	fail 0.889 / 35.002
PatchTST	60	fail 0.500 / 35.035	fail 1.667 / 34.342	fail 2.222 / 35.674	fail 2.333 / 35.133

Each entry gives PTE in weeks / PIE in percent. s0–s3 are the first to fourth forecast weeks. The detector uses prominence 0.15 × range, distance 15, and ±8-week matching. These detector settings and joint aggregation are project choices. Source: BENCH_PTE_PIE_BY_STEP.csv.

F1 conclusion. NLinear and MIFlu each pass 4/16 aggregated cells, while PatchTST passes 0/16. NLinear has lower mean PIE (9.477% versus MIFlu 26.846%) but higher mean PTE (1.677 versus 1.128

weeks). This is a timing–magnitude trade-off, not dominance. The ordering is inverted at both ends among the three leaders: MSE orders MIFlu, PatchTST, NLinear; PIE orders NLinear, MIFlu, PatchTST.



Step-zero mean PTE and PIE across horizons for the three leading arms. Mean-level tolerances are illustrative; MIFlu still fails L48 at step zero. A mean error below a threshold is not a season-level success rate. Source: BENCH_PTE_PIE_BY_STEP.csv.

Table 6. Parameter analysis and rank associations

Quantity	Value
Regression slope / R^2	+0.1171 / 0.0223
Models with parameter exports	11
Subset Pareto set	NLinear 2,520; PatchTST 531,878
Parameter ratio	211.1×
Descriptive MSE reduction	11.2%
ρ(MSE, PIE)	+0.736 ($p < 0.0001$)
ρ(MSE, passes)	-0.429 ($p = 0.023$)

The Pareto analysis includes only 11 arms with parameter exports. MIFlu and the frozen encoders are not in that subset. Missing parameter counts are not zero. Parameter count is not a measurement of latency, memory or energy. Correlations are descriptive across models on one test set.

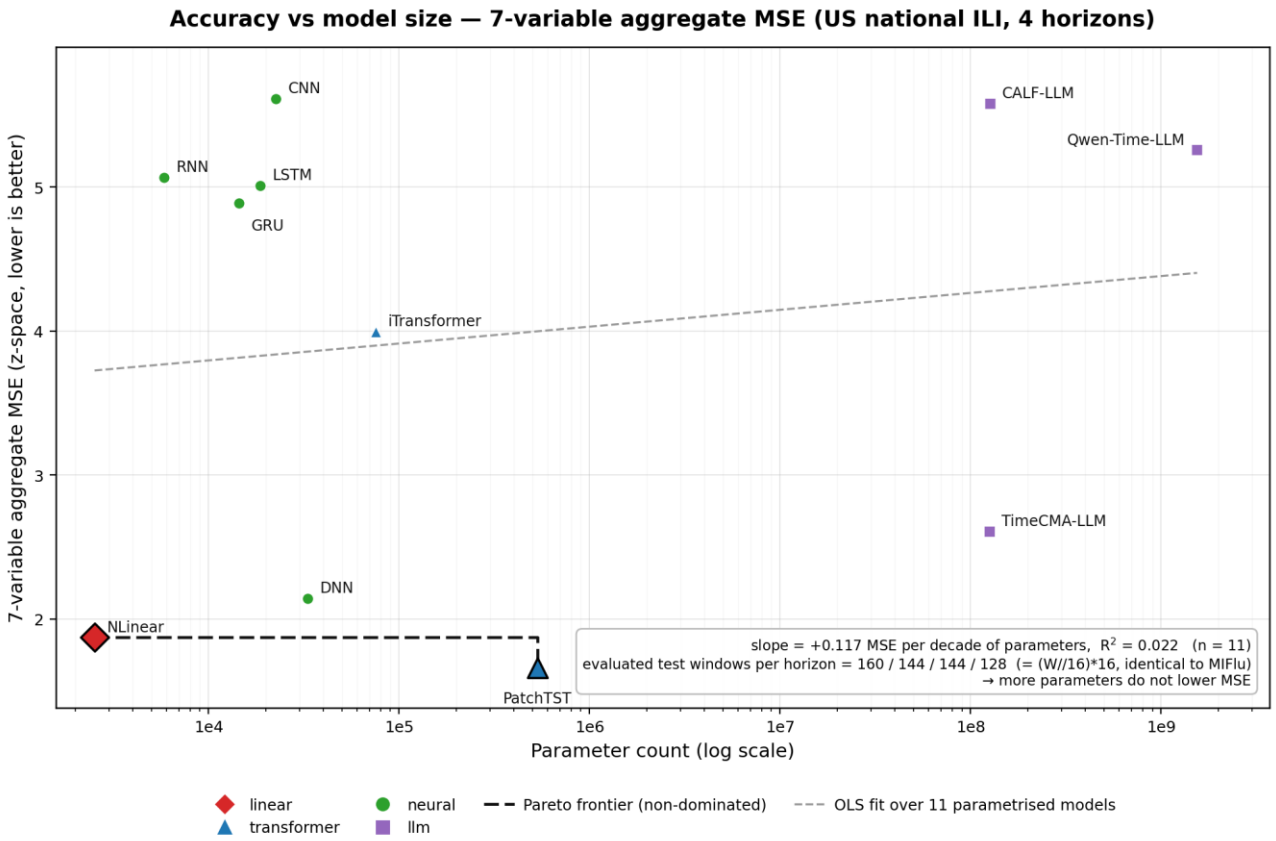
F2 conclusion. The fitted slope is +0.1171 and R^2 is 0.0223 across the 11 arms with exported parameter counts. Size alone explains little of their observed MSE variation. This subset result does not prove that capacity is irrelevant.

Table 7. Paired comparisons with corrected Holm adjustment

Model A	Model B	Δ MSE	p raw	p Holm	Effect*
NLinear	PatchTST	+0.2095	0.005859	0.1172	1.15
Gradient Boosting	RNN	-1.4029	0.000488	0.0664	4.86
KNN	NLinear	+1.9138	0.000488	0.0664	22.29
KNN	MIFlu	+2.2280	0.000488	0.0664	9.89
KNN	LSTM	-1.2234	0.000488	0.0664	5.28
iTransformer	TimeCMA	+1.3863	0.000488	0.0664	7.02
iTransformer	SVR	-1.3443	0.000488	0.0664	10.61

*136 pairs cover 17 models, not all 28. n = 12 matched horizon × seed cells. Forward cumulative-max Holm yields 0/136 rejections. The shown p values recover the exact 1/4096 enumeration grid from six-decimal exports; recomputing directly from rounded exports gives NLinear–PatchTST 0.117180 instead of 0.1171875. *The source effect uses pooled model SD, not paired-difference SD. Cell dependence limits inference.*

F3 conclusion. Proper Holm gives 0/136 rejections (128/136 before adjustment), with a minimum adjusted p of about 0.0664 and NLinear–PatchTST p ≈ 0.1172. PatchTST still has 11.2% lower observed MSE with 211× the parameters and lower MSE in 10/12 cells. Non-rejection does not establish equivalence or show that the gain is noise.



Accuracy–parameter trade-off among 11 models with exported parameter counts. MIFlu and frozen encoders are omitted from this subset. Parameter count is not a measured runtime or memory cost. Source: BENCH_EFFICIENCY.csv.

Table 8. Frozen encoders with shared linear readout

Encoder	MSE	vs MIFlu	PTE	PIE (%)	Passes
Hunyuan-1.8B	4.7374	3.04×	4.406	71.428	0/16
Hunyuan-7B	4.7596	3.06×	5.309	75.399	0/16
DeepSeek-R1-Qwen-1.5B	4.7866	3.07×	5.792	78.061	0/16
DeepSeek-R1-Llama-8B	4.8250	3.10×	4.306	77.121	0/16
DeepSeek-R1-32B	4.9372	3.17×	3.889	77.730	0/16
DeepSeek-R1-Qwen-14B	5.0166	3.22×	4.149	76.350	0/16
DeepSeek-R1-Qwen-7B	5.0308	3.23×	4.458	76.373	0/16
ContextPilot-8B	5.0843	3.26×	4.962	78.918	0/16
Hunyuan-4B	5.1317	3.30×	3.003	78.840	0/16
ContextPilot-14B	5.1949	3.34×	4.965	77.764	0/16
Qwen2.5-1.5B	5.2049	3.34×	5.333	74.650	0/16

11 retained encoders × 4 horizons × 3 seeds = 132 cells. The historical plan included two additional embedding-only arms, giving 156 planned cells; those 24 omitted cells are outside the 28-model roster. The result tests only this frozen-feature pipeline.

F4 conclusion. The 11 tested frozen encoders score 4.7374–5.2049, or 3.04–3.34× MIFlu's MSE. All exceed the decision tree's 4.3198 and pass 0/16 cells. This is a negative result for this feature/readout configuration; prompting, fine-tuning and numerical foundation models remain untested.

Table 9. Ablation design and snapshot status

Variant	Flag	Cells	Status
Reference K=6	default	12	Already in benchmark
no_lora	--no-lora	12	DONE (not aggregated)
K=2	--num-layers 2	12	DONE (not aggregated)
K=4	--num-layers 4	12	DONE (not aggregated)
K=8	--num-layers 8	12	DONE (not aggregated)
K=12	--num-layers 12	12	DONE (not aggregated)

Returned Burgundy queue evidence dated 4 November 2026 records 60/60 additional ablation cells DONE. Per-variant seven-channel aggregation remains pending; this review did not independently rerun the ablation jobs. Historical job 722968 is reported CANCELLED, not active. The baseline 12 cells are not part of the additional 60. No trend is inferred.

F5 provisional conclusion. In miflu_model.py's author_faithful path, text follows time tokens behind causal attention and the head reads the time tokens. Static inspection therefore predicts no text influence during deterministic evaluation. The supplied tests_archive/test_miflu_ablations.py explicitly asserts both text invariance and full/no-text equality with matched CPU initialisation. The supplied JUnit and runtime log now record three no-text assertions passed with zero skips under fixed-weight CPU evaluation; they were not rerun in this review. Other fusion paths and training randomness require separate checks.

Problems encountered. The source exported Holm column remains defective; the corrected corpus recomputes the proper adjustment. The roster arithmetic, MAE crossover, hardware coverage and missing tests also needed correction. Queue status cannot be certified from this offline snapshot. I will freeze the corrected reporting path and keep all original CSVs for comparison.

7. Summary of objectives addressed and results achieved to date

Table 10. Objectives addressed and estimated completion

Objective	Addressed	Estimated completion
Reproduction arithmetic	Yes, source provenance limited	100%
28-arm benchmark export	Yes, arithmetic verified	100%
Peak and efficiency evaluation	Yes, interpretation corrected	100%
Frozen encoder axis	Yes, retained arms only	100%
Ablation	Queue DONE 60/60; analysis pending	100% queue only
Rolling origin	Code stage, not evaluated	15%
Manuscript	Draft; publication evidence incomplete	70%

The ablation queue completion is 100% (60/60), while its per-variant analysis remains pending. Other project-completion percentages are owner planning estimates, not measured research scores. The old statement equating 45% with 60/60 was incorrect.

8. Work schedule and expected results for the next one to two months

Table 11. Work schedule and expected outputs

Target date	Task	Expected output and fallback
By 16 Nov 2026	Pull and verify ablation exports	Measured table if complete; otherwise retain incomplete label
16–20 Nov 2026	Present to supervisor and assessor	Appointment and attendance must be confirmed by student
20 Nov 2026 23:59 HKT	Submit through FYP InfoNet	Submission receipt; late penalties apply
By 30 Nov 2026	Regional table and timing provenance	Complete denominators and hardware details; no assumed trend
By 31 Dec 2026	Rolling origins and detector sensitivity	Season-level uncertainty and full-window sensitivity
Jan–Apr 2027	Publication decision	JBHI-oriented draft only if evidence and scope are adequate

Dates are targets. Queue availability, measured runtime and supervisor approval determine feasibility. No conference deadline is inferred from this schedule.

Risks and mitigation. Queue delays may prevent ablation completion, so retain explicit incomplete labels. Validate the date mapping and use season-level rolling origins to address temporal dependence. Restore the original tests and pin dependencies before publication. Confirm a presentation appointment with both supervisor and assessor. The school template requires both presentation and timely InfoNet submission.

Date _____ Signature of student _____

Key references used in the background and methodology are listed below. Numbering matches the internal report and manuscript.

[1] Jaeuk Moon, Jonghwa Shim, Eunbeen Kim and Eenjun Hwang, "MIFlu: Large Language Model-Based Multimodal Influenza Forecasting Scheme," IEEE JBHI 29(10), 7790-7801, 2025. https://doi.org/10.1109/JBHI.2025.3561214

[2] Sarah C. Kramer and Jeffrey Shaman, "Development and validation of influenza forecasting for 64 temperate and tropical countries," PLOS Computational Biology 15(2), e1006742, 2019. https://doi.org/10.1371/journal.pcbi.1006742

[9] Abdul Fatir Ansari et al., "Chronos: Learning the Language of Time Series," Transactions on Machine Learning Research, 2024. https://arxiv.org/abs/2403.07815

[13] Mingtian Tan, Mike A. Merrill, Vinayak Gupta, Tim Althoff and Thomas Hartvigsen, "Are Language Models Actually Useful for Time Series Forecasting?," NeurIPS 37, 2024. https://arxiv.org/abs/2406.16964

[14] Xin Qiu et al., "Rethinking the Role of LLMs in Time Series Forecasting," arXiv:2602.14744 v1, preprint, 2026. https://arxiv.org/abs/2602.14744

[18] Johannes Bracher, Evan L. Ray, Tilmann Gneiting and Nicholas G. Reich, "Evaluating epidemic forecasts in an interval format," PLOS Computational Biology 17(2), e1008618, 2021. https://doi.org/10.1371/journal.pcbi.1008618

V8 evidence update (2026-11-07)

The verified national common panel is complete: 1,120/1,120 cells across 28 arms, four horizons, and ten seeds (2021-2030). All-valid MSE means are MIFlu 1.7944, PatchTST 1.9521, and NLinear 2.0782; the crop panel is 1.4994, 1.6352, and 1.9027 respectively.

The legacy five-fold walk-forward panel is a separate population and is complete at 1,020/1,020 valid cells (seeds 42/43/44), with MIFlu 2.2203, PatchTST 3.0396, and NLinear 3.2116. It must not be pooled with the national panel.

Foundation models are a separate zero-shot roster: Chronos and TimesFM have 80/80 valid rolling-origin cells (ten origins, not training seeds). Their univariate per-channel adapter discards cross-channel dependence, so they are not part of the 28-arm or 1,120-cell comparison.

The ablation is non-monotonic (K4 1.4989 best; K12 1.7604 worst). Resource evidence covers six representative models, not a full-1,120-cell Pareto analysis. CDC vintage analysis is a retrospective final-vintage reconstruction, not real-time validation.

Risks and remaining verification items: the literal foundation 10-seed expansion is not executed by protocol; the information-set-matched multivariate foundation control, dependence-aware confirmatory inference, isolated MIFlu/NLinear/PatchTST forward latency, full-panel resource Pareto, and exact JBHI template class remain blocked or unverified. Validator PASS is structural only (scientific_readiness: NOT_ASSESSED).